



SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL
SCIENCES



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COURSE: CSA0915 -JAVA PROGRAMMING

SLOT: B

1. The following Java program is intended to print all elements of an array. However, it throws an error during execution. Identify and correct the mistake in the loop condition.

The screenshot shows the SIMATS Online Programming Lab interface. On the left, there is a sidebar with a list of questions (Q5 to Q10) and their respective marks. The main area is divided into three sections: 'Your Code', 'Input', and 'Output'. The 'Your Code' section contains the following Java code:

```
1 import java.util.*;
2 class ArrayIndex
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner (System.in);
7         int []arr = new int[6];
8         for(int i=0; i<arr.length;i++)
9         {
10             arr[i] = sc.nextInt();
11         }
12
13         for(int i = 0; i < arr.length; i++){
14             System.out.println(arr[i]);
15         }
16     }
17 }
18
19
```

The 'Input' section shows the input: 2 3 4 5 6 7. The 'Output' section shows the output: 2 3 4 5 6 7. The 'Visible Test Cases' section shows three test cases: Test Case 1, Test Case 2, and Test Case 3. The 'Run' button is highlighted.

2. A program attempts to print the length of a string but results in a runtime exception. Analyze the issue and modify the code to handle it properly.

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Your Code

```
1 import java.util.*;
2 class Q2_NullPointer {
3     public static void main(String[] args)
4     {
5         String str = "Lochana";
6
7         if (str != null) {
8             System.out.println(str.length());
9         } else {
10            System.out.println("String is null");
11        }
12    }
13 }
14
```

Run

Save

Input

stdin input

Output

7

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

3. A developer compares two strings using the `==` operator, which leads to incorrect output. Identify the issue and correct the comparison method.

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Your Code

```
1 import java .util.*;
2 class Q3_StringCompare {
3     public static void main(String[] args) {
4         String s1 = "Java";
5         Scanner sc = new Scanner (System.in);
6         String s2 = sc.nextLine();
7
8         if (s1.equals(s2))
9         {
10            System.out.println("Equal");
11        }
12        else
13        {
14            System.out.println("Not Equal");
15        }
16    }
17 }
18
```

Run

Save

Input

Java

Output

Equal

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

4. The given program uses a loop to print numbers from 0 to 4, but it runs infinitely. Identify the cause and fix the loop.

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Your Code

```
1 import java.util.*;
2 class Q4_Loop
3 {
4     public static void main(String[] args)
5     {
6         int i = 0;
7         while(i < 5){
8             System.out.println(i);
9             i++;
10        }
11    }
12 }
```

Input

stdin input

Output

```
0
1
2
3
4
```

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

Run Save

5. A program is written to reverse a string. Verify whether the logic is correct and modify it if necessary.

}

Your Code

```

1 import java.util.*;
2 class Q5_Reverse
3 {
4     public static void main(String[] args)
5     {
6         String str = "Hello";
7         String rev = "";
8         for(int i = 0; i < str.length(); i++)
9         {
10             rev = str.charAt(i) + rev;
11         }
12 System.out.println(rev);
13     }
14 }

```

Run

Save

Input

stdin input

Output

olleH

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

6. The program calculates the sum of array elements, but it produces an incorrect result. Identify the problem and fix it.

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}

Your Code

```

1 import java.util.*;
2 class Q6_ArraySum
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int arr[] = new int[5];
8         for(int i=0; i<arr.length;i++)
9         {
10             arr[i]=sc.nextInt();
11         }
12         int sum = 0;
13         for(int i=0;i<arr.length;i++)
14         {
15             sum += arr[i];
16         }
17 System.out.println(sum);
18     }
19 }

```

Run

Save

Input

5 10 15 20 25

Output

75

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

7. A program tries to access a character from a string using an invalid index, causing an exception. Identify and correct the issue.

Your Code

```
1 import java.util.*;
2 class Q7_StringIndex
3 {
4     public static void main(String[] args)
5     {
6         String str = "Java";
7         System.out.println(str.charAt(3));
8     }
9 }
```

Input

stdin input

Output

a

Visible Test Cases

8. A string comparison fails due to case sensitivity. Modify the program to make the comparison case-insensitive.

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```
}

```

Your Code

```
1 import java.util.*;
2 class Q8_CaseInsensitive
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         String str = sc.nextLine();
8         if(str.equalsIgnoreCase("JAVA"))
9         {
10             System.out.println("Match");
11         }
12         else
13         {
14             System.out.println("Not Match");
15         }
16     }
17 }
```

Input

jAvA

Output

Match

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

Run

Save

9. The program contains an error in array declaration and initialization. Identify the syntax issue and correct it.

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```
}  
}
```

Your Code

```
1 import java.util.Arrays;  
2  
3 class Q9_ArrayDeclaration  
4 {  
5     public static void main(String[] args)  
6     {  
7         int arr[] = {1,2,3};  
8         System.out.println(Arrays.toString(arr));  
9     }  
10 }
```

Input
stdin input

Output
[1, 2, 3]

Visible Test Cases

10. A switch statement prints multiple outputs due to missing statements. Identify the problem and correct the code.

Your Code

```
1 import java.util.*;
2 class Q10_Switch
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int num = sc.nextInt();
8         switch(num)
9         {
10             case 1:
11                 System.out.println("One");
12                 break;
13             case 2:
14                 System.out.println("Two");
15                 break;
16             case 3:
17                 System.out.println("Three");
18                 break;
19             default:
20                 System.out.println("Invalid Input");
21         }
22     }
23 }
```

 Run

 Save

Input

3

Output

Three

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3